Unit Test Report

for

Student Attendance Management System (SAMS)

Version: 1.0

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Team 1

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1. Introduction

This report gives the results of unit testing of SAMS, done as per section 5 of the Software Test Plan. Unit tests are written with Django TestCase and cover models, views, forms and helper functions of each Django app. Every test is mapped to the SRS requirement and the Test Plan test case it checks.

2. Test Environment

Item	Details
Code under test	main branch at commit c1edd6d (after PR #39) + tests added in this testing week
Language / framework	Python 3.13, Django 4.2
Database	SQLite in-memory test database (USE_SQLITE=1), same as GitHub Actions CI
Test runner	Django test runner (unittest)
Command	cd sams USE_SQLITE=1 python manage.py test -v 2
Date run	02-11-2026

3. Summary

Total tests	Passed	Failed	Errors	Skipped	Time
121	121	0	0	0	44.7 s

Django app	Tests	Passed
academics	6	6
accounts	19	19
attendance	27	27
leaves	34	34
reports	35	35
Total	121	121

All 121 unit tests passed. 93 tests existed at the end of Sprint 2; 28 tests were added during the testing week (marked "new" in section 4) to cover validation paths that the coverage run showed were untested.

4. Results by Test Class

App	Test class	Tests	SRS req.	Test Plan TC	Result
academics	AcademicsAdminFormTests (new)	3	REQ-6, 7	TC-05	Pass
academics	AcademicsConstraintTests	3	REQ-5, 6, 7	TC-05	Pass
accounts	AdminRoleTests	3	REQ-3, NFR-7	TC-05	Pass
accounts	DashboardTests	1	REQ-2	TC-03	Pass
accounts	ImporterValidationTests (new)	4	REQ-4	TC-04	Pass
accounts	LoginTests	2	REQ-1	TC-01, TC-02	Pass
accounts	RoleRequiredDecoratorTests	2	NFR-7	TC-03	Pass
accounts	StudentExcelUploadTests	7	REQ-4	TC-04	Pass
attendance	AttendancePercentTests	4	REQ-11, BR-5	TC-09	Pass
attendance	EditAttendanceTests	8	REQ-10, NFR-5, NFR-11	TC-08	Pass
attendance	MarkAttendanceTests	6	REQ-8, 9	TC-06, TC-07	Pass
attendance	ShortageHookTests	9	REQ-11, 16	TC-13	Pass
leaves	ApplyLeaveTests	7	REQ-17, NFR-8	TC-14	Pass
leaves	ApproveLeaveTests	8	REQ-18, 19, NFR-11	TC-14	Pass
leaves	DecideLeaveBadInputTests (new)	10	REQ-18	TC-14	Pass
leaves	LeaveValidationTests (new)	3	REQ-17	TC-14	Pass
leaves	ShortageBoundaryTests	1	REQ-16, BR-4	TC-13	Pass
leaves	ShortageEmailTests	5	REQ-16	TC-13	Pass
reports	ClassReportTests	6	REQ-13	TC-11	Pass
reports	MyAttendanceTests	4	REQ-12, NFR-8	TC-10	Pass

reports	ReportExportTests	8	REQ-14	TC-11	Pass
reports	ShortageFilterTests (new)	8	REQ-15	TC-12	Pass
reports	ShortageListTests	6	REQ-15, BR-4	TC-12	Pass
reports	IsBelowTests	3	BR-4	TC-12	Pass

Note: some classes reuse the setup of another class by inheritance (for example DecideLeaveBadInputTests extends ApproveLeaveTests), so the inherited tests run again inside them. These count in the total of 121.

5. Defects Found and Fixed

ID	Found by	Defect	Fix	Status
D-1	Code review	createsuperuser created a user with role "student", so the first admin landed on the student dashboard	User.save() sets role admin for superusers (PR #36)	Fixed, retested
D-2	Code review	Users with role "admin" could not open the Django admin site (not staff), so REQ-3, REQ-5 to 7 could not be done	Admin role users get staff access (PR #36)	Fixed, retested
D-3	Code review	Shortage email used the rounded % while the shortage list used the exact value, so they could disagree at the 75% boundary	Both use exact counts (PR #36)	Fixed, retested
D-4	Code review	Shortage list showed % across all classes of the subject but decided "below 75" per class	% calculated per class (PR #36)	Fixed, retested

No open defects at the end of unit testing.

Appendix A: Full Test List

#	App	Test class	Test	Result
1	academics	AcademicsAdminFormTests	test_class_form_lists_only_students	Pass
2	academics	AcademicsAdminFormTests	test_class_list_shows_student_count	Pass
3	academics	AcademicsAdminFormTests	test_teaching_form_lists_only_faculty	Pass

4	academics	AcademicsConstraintTests	test_class_unique	Pass
5	academics	AcademicsConstraintTests	test_subject_code_unique	Pass
6	academics	AcademicsConstraintTests	test_teaching_unique_on_subject_and_class	Pass
7	accounts	AdminRoleTests	test_admin_role_can_open_django_admin	Pass
8	accounts	AdminRoleTests	test_createsuperuser_gets_admin_role	Pass
9	accounts	AdminRoleTests	test_faculty_and_student_are_not_staff	Pass
10	accounts	DashboardTests	test_each_role_gets_its_own_template	Pass
11	accounts	ImporterValidationTests	test_excel_floats_become_ints	Pass
12	accounts	ImporterValidationTests	test_existing_username_rejected	Pass
13	accounts	ImporterValidationTests	test_field_rules	Pass
14	accounts	ImporterValidationTests	test_too_many_rows	Pass
15	accounts	LoginTests	test_valid_login_redirects_to_dashboard	Pass
16	accounts	LoginTests	test_wrong_password_shows_invalid_credentials_message	Pass
17	accounts	RoleRequiredDecoratorTests	test_matching_role_is_allowed	Pass
18	accounts	RoleRequiredDecoratorTests	test_wrong_role_is_forbidden	Pass
19	accounts	StudentExcelUploadTests	test_admin_only	Pass
20	accounts	StudentExcelUploadTests	test_blank_rows_ignored	Pass
21	accounts	StudentExcelUploadTests	test_no_file	Pass
22	accounts	StudentExcelUploadTests	test_non_xlsx_rejected	Pass
23	accounts	StudentExcelUploadTests	test_rejected_rows_listed_and_valid_rows_still_saved	Pass
24	accounts	StudentExcelUploadTests	test_valid_rows_create_students_and_class	Pass
25	accounts	StudentExcelUploadTests	test_wrong_header_rejected	Pass
26	attendance	AttendancePercentTests	test_leave_counts_as_attended	Pass
27	attendance	AttendancePercentTests	test_present_over_total	Pass
28	attendance	AttendancePercentTests	test_scoped_to_subject_and_student	Pass
29	attendance	AttendancePercentTests	test_zero_when_no_classes	Pass
30	attendance	EditAttendanceTests	test_admin_can_edit_after_3_days	Pass
31	attendance	EditAttendanceTests	test_faculty_blocked_after_3_days	Pass

32	attendance	EditAttendanceTests	test_faculty_edit_within_3_days_logs_change	Pass
33	attendance	EditAttendanceTests	test_invalid_status_ignored_and_nothing_deleted	Pass
34	attendance	EditAttendanceTests	test_other_faculty_cannot_edit	Pass
35	attendance	EditAttendanceTests	test_students_cannot_edit	Pass
36	attendance	EditAttendanceTests	test_unchanged_statuses_write_no_log	Pass
37	attendance	EditAttendanceTests	test_unmarked_slot_shows_message	Pass
38	attendance	MarkAttendanceTests	test_default_present_and_ticked_absent	Pass
39	attendance	MarkAttendanceTests	test_duplicate_is_rejected	Pass
40	attendance	MarkAttendanceTests	test_future_date_rejected	Pass
41	attendance	MarkAttendanceTests	test_lists_all_students	Pass
42	attendance	MarkAttendanceTests	test_only_assigned_faculty	Pass
43	attendance	MarkAttendanceTests	test_requires_login_and_faculty_role	Pass
44	attendance	ShortageHookTests	test_default_present_and_ticked_absent	Pass
45	attendance	ShortageHookTests	test_duplicate_is_rejected	Pass
46	attendance	ShortageHookTests	test_future_date_rejected	Pass
47	attendance	ShortageHookTests	test_hook_runs_per_student_after_save	Pass
48	attendance	ShortageHookTests	test_lists_all_students	Pass
49	attendance	ShortageHookTests	test_mail_failure_does_not_break_saving	Pass
50	attendance	ShortageHookTests	test_missing_check_shortage_is_noop	Pass
51	attendance	ShortageHookTests	test_only_assigned_faculty	Pass
52	attendance	ShortageHookTests	test_requires_login_and_faculty_role	Pass
53	leaves	ApplyLeaveTests	test_faculty_cannot_apply	Pass
54	leaves	ApplyLeaveTests	test_login_required	Pass
55	leaves	ApplyLeaveTests	test_overlapping_request_rejected	Pass
56	leaves	ApplyLeaveTests	test_rejected_request_does_not_block_new_one	Pass
57	leaves	ApplyLeaveTests	test_student_can_apply_and_request_is_pending	Pass
58	leaves	ApplyLeaveTests	test_student_sees_only_own_requests	Pass
59	leaves	ApplyLeaveTests	test_to_date_before_from_date_rejected	Pass

60	leaves	ApproveLeaveTests	test_approve_marks_absent_days_in_range_as_leave	Pass
61	leaves	ApproveLeaveTests	test_approved_leave_counts_in_percentage	Pass
62	leaves	ApproveLeaveTests	test_cannot_decide_twice	Pass
63	leaves	ApproveLeaveTests	test_faculty_sees_pending_request	Pass
64	leaves	ApproveLeaveTests	test_get_not_allowed	Pass
65	leaves	ApproveLeaveTests	test_other_faculty_cannot_decide	Pass
66	leaves	ApproveLeaveTests	test_reject_leaves_records_alone	Pass
67	leaves	ApproveLeaveTests	test_student_cannot_decide	Pass
68	leaves	DecideLeaveBadInputTests	test_approve_marks_absent_days_in_range_as_leave	Pass
69	leaves	DecideLeaveBadInputTests	test_approved_leave_counts_in_percentage	Pass
70	leaves	DecideLeaveBadInputTests	test_cannot_decide_twice	Pass
71	leaves	DecideLeaveBadInputTests	test_faculty_sees_pending_request	Pass
72	leaves	DecideLeaveBadInputTests	test_get_not_allowed	Pass
73	leaves	DecideLeaveBadInputTests	test_invalid_decision_changes_nothing	Pass
74	leaves	DecideLeaveBadInputTests	test_no_email_address_still_decides	Pass
75	leaves	DecideLeaveBadInputTests	test_other_faculty_cannot_decide	Pass
76	leaves	DecideLeaveBadInputTests	test_reject_leaves_records_alone	Pass
77	leaves	DecideLeaveBadInputTests	test_student_cannot_decide	Pass
78	leaves	LeaveValidationTests	test_model_str	Pass
79	leaves	LeaveValidationTests	test_more_than_30_days_rejected	Pass
80	leaves	LeaveValidationTests	test_short_reason_rejected	Pass
81	leaves	ShortageBoundaryTests	test_mail_once_it_drops_below_75	Pass
82	leaves	ShortageEmailTests	test_mail_sent_when_below_75	Pass
83	leaves	ShortageEmailTests	test_marking_attendance_triggers_mail	Pass
84	leaves	ShortageEmailTests	test_no_email_address_no_mail	Pass
85	leaves	ShortageEmailTests	test_no_mail_at_exactly_75	Pass
86	leaves	ShortageEmailTests	test_not_resent_within_a_week	Pass
87	reports	ClassReportTests	test_anonymous_redirected_to_login	Pass

88	reports	ClassReportTests	test_date_range_limits_records	Pass
89	reports	ClassReportTests	test_faculty_cannot_report_on_another_facultys_class	Pass
90	reports	ClassReportTests	test_faculty_sees_appendix_b_columns	Pass
91	reports	ClassReportTests	test_start_after_end_is_rejected	Pass
92	reports	ClassReportTests	test_student_forbidden	Pass
93	reports	MyAttendanceTests	test_74_96_percent_is_flagged_not_rounded_up_to_75	Pass
94	reports	MyAttendanceTests	test_faculty_forbidden_and_anonymous_redirected	Pass
95	reports	MyAttendanceTests	test_leave_counts_as_attended	Pass
96	reports	MyAttendanceTests	test_student_sees_only_own_attendance	Pass
97	reports	ReportExportTests	test_anonymous_redirected_to_login	Pass
98	reports	ReportExportTests	test_excel_download_has_appendix_b_fields_and_rows	Pass
99	reports	ReportExportTests	test_faculty_cannot_export_another_facultys_class	Pass
100	reports	ReportExportTests	test_invalid_range_rejected	Pass
101	reports	ReportExportTests	test_pdf_download	Pass
102	reports	ReportExportTests	test_report_page_shows_download_buttons	Pass
103	reports	ReportExportTests	test_student_forbidden	Pass
104	reports	ReportExportTests	test_unknown_format_is_404	Pass
105	reports	ShortageFilterTests	test_admin_sees_all_classes	Pass
106	reports	ShortageFilterTests	test_anonymous_redirected_to_login	Pass
107	reports	ShortageFilterTests	test_email_sent_column	Pass
108	reports	ShortageFilterTests	test_filter_by_one_class	Pass
109	reports	ShortageFilterTests	test_lists_only_students_below_75	Pass
110	reports	ShortageFilterTests	test_model_strings	Pass
111	reports	ShortageFilterTests	test_other_faculty_sees_nothing	Pass
112	reports	ShortageFilterTests	test_student_forbidden	Pass
113	reports	ShortageListTests	test_admin_sees_all_classes	Pass
114	reports	ShortageListTests	test_anonymous_redirected_to_login	Pass
115	reports	ShortageListTests	test_email_sent_column	Pass

116	reports	ShortageListTests	test_lists_only_students_below_75	Pass
117	reports	ShortageListTests	test_other_faculty_sees_nothing	Pass
118	reports	ShortageListTests	test_student_forbidden	Pass
119	reports	IsBelowTests	test_above_75_and_no_classes	Pass
120	reports	IsBelowTests	test_exactly_75_is_not_below	Pass
121	reports	IsBelowTests	test_just_under_75_is_below	Pass